

VoxHalo

Vocal Chain + Hybrid Echo

User Manual | Version 1.0 | macOS (Audio Unit, VST3, Standalone)

Welcome to VoxHalo

VoxHalo is a complete vocal chain in one window. It cleans, compresses, warms and brightens a voice, then adds a modulated stereo echo and a soft halo reverb. Every control is a vertical slider, grouped into colour-coded sections so you can read the whole chain at a glance.

The plug-in is built for lead vocals, ad-libs, loop-station layers and spoken word. Twenty factory presets cover jazz, soul, hip-hop, trap, afrobeats and ambient sounds, and every preset is a starting point you can reshape.

Highlights

- **One window, six sections:** Input + Clean, Squash, Glow, Echo, Halo and Output.
- **Hybrid echo:** free time or tempo sync, ping-pong spread, filters inside the feedback loop and tape-style modulation.
- **Resizable and sharp:** drag the bottom-right corner to scale the interface from half size to double size.
- **Tooltips everywhere:** hover any control for a plain-language explanation.
- **20 presets** reachable from the preset menu, the arrow buttons or your DAW's own preset list.

Interface map



The map above is a schematic. Sections run left to right in signal order, and the Echo section carries the Sync switch and note-length menu on its title row.

Installation (macOS)

System requirements: macOS 11 or later, Apple Silicon or Intel (universal build). A host that supports Audio Unit or VST3 plug-ins, such as Logic Pro, GarageBand, Ableton Live, Reaper or Studio One. VoxHalo is a stereo plug-in.

Using the installer

- Open **VoxHalo-1.0.0.dmg** and double-click **VoxHalo-1.0.0.pkg**.
- Follow the prompts. Enter your Mac password when asked.
- The installer places the files shown in the table below.
- Restart your DAW. Logic Pro scans new Audio Units automatically on launch.

File	Location
VoxHalo.component (Audio Unit)	/Library/Audio/Plug-Ins/Components
VoxHalo.vst3	/Library/Audio/Plug-Ins/VST3
VoxHalo.app (Standalone)	/Applications
This manual (PDF)	/Library/Application Support/VoxHalo

First launch and Gatekeeper

Builds that are not notarized by Apple show a warning the first time. Control-click (or right-click) the installer, choose Open, then confirm. If macOS still blocks a plug-in, open System Settings, Privacy and Security, and choose Open Anyway next to the VoxHalo message.

Using VoxHalo in your DAW

- **Logic Pro:** add an Audio FX slot, choose Audio Units, then VoxHalo Audio, then VoxHalo. If it does not appear, open Logic Pro, Settings, Plug-in Manager and press Reset and Rescan Selection.
- **Ableton Live:** enable VST3 and Audio Units in Preferences, Plug-ins, then rescan.
- **Standalone app:** open VoxHalo from Applications, pick your audio interface in the settings button, and sing. Use headphones to avoid feedback.

Uninstalling

Delete the three items listed in the table above (the plug-in files and the app). No other files are installed.

Quick start

- **1. Pick a preset.** Use the menu at the top or the < and > buttons. Start with Init Clean or Velvet Lead.
- **2. Set Input.** Sing a loud phrase and lower Input until the signal stays comfortably away from clipping.
- **3. Shape the voice.** Raise Low Cut until the boom disappears, add De-Ess if S sounds sharp, then bring up Squash until the level feels steady.
- **4. Add colour.** A little Drive adds warmth. Air lifts the top end or softens a bright microphone.
- **5. Add space.** Turn Echo Mix up for repeats and Halo Mix up for room. Turn on Sync to lock the echo to your tempo.
- **6. Balance.** Use Mix to blend with the dry voice and Output to set the final level.

Handy tips

- Double-click any slider to return it to its default value.
- Click a value under a slider to type an exact number. Typing 2k sets 2 kHz.
- Hold Shift while dragging for fine control.
- Hover over any control for a tooltip.
- Resize the window by dragging the bottom-right corner.
- Presets load instantly and are saved with your DAW project, along with any tweaks you made.

Signal flow



Dry input is tapped after the Input knob and blended back in by the Mix slider. Reverb listens to the post-echo signal.

Processing runs in the order shown. Cleaning comes first so rumble and harsh sibilance never reach the compressor or the echo. The echo feeds the reverb, so every repeat blooms into the room.

Control reference

INPUT + CLEAN

Control	Range	Default	What it does
Input	-24 to +12 dB	0 dB	Gain into the chain. Set it so peaks stay clear of clipping.
Low Cut	20 to 400 Hz	80 Hz	High-pass filter that removes rumble, plosives and proximity boom.
De-Ess	0 to 100%	30%	Reduces harsh S and T sounds above about 6.5 kHz.

SQUASH

Control	Range	Default	What it does
Squash	0 to 100%	40%	One-knob compressor. Higher settings lower the threshold, raise the ratio and add automatic makeup gain.
Attack	0.1 to 80 ms	8 ms	How quickly compression starts. Slower keeps more punch.
Release	20 to 800 ms	160 ms	How quickly compression lets go. Longer is smoother.

GLOW

Control	Range	Default	What it does
Drive	0 to 100%	15%	Warm saturation that adds harmonics and presence.
Air	-6 to +9 dB	+2 dB	High shelf at 9 kHz. Positive values add sparkle, negative values soften.

ECHO

Control	Range	Default	What it does
Time	20 to 1500 ms	320 ms	Delay time. Greyed out while Sync is on.
Feedback	0 to 95%	35%	Number of repeats. A safety limiter keeps high settings under control.
Hi Cut	1 to 20 kHz	6 kHz	Low-pass filter in the feedback loop. Each repeat grows darker.
Lo Cut	20 to 1000 Hz	150 Hz	High-pass filter in the feedback loop. Keeps repeats light and clear of the vocal.
Mod Rate	0.05 to 8 Hz	0.6 Hz	Speed of the delay-time modulation.
Mod Depth	0 to 100%	15%	Amount of modulation, up to plus or minus 8 ms. Opposite phase on left and right adds width.
Spread	0 to 100%	50%	Stereo ping-pong. At 100% the echoes alternate fully between left and right.
Echo Mix	0 to 100%	25%	Level of the echo added to the vocal.
Sync / Division	Off, 1/16 to 1/1	Off, 1/4	Locks the echo to the DAW tempo. T is triplet and D is dotted. Without tempo information the plug-in assumes 120 BPM.

HALO

Control	Range	Default	What it does
Size	0 to 100%	40%	Room size, from a small booth to a cathedral.
Damp	0 to 100%	50%	High-frequency absorption. Higher is darker.
Halo Mix	0 to 100%	15%	Reverb level.

OUTPUT

Control	Range	Default	What it does
Mix	0 to 100%	100%	Blend between the dry input and the full chain.
Output	-24 to +12 dB	0 dB	Final level.

Factory presets

Twenty presets ship with VoxHalo. Names in the Echo column show the tempo-synced note length where Sync is on.

#	Name	Echo	Character
1	**Init Clean**	320 ms	Balanced starting point with gentle polish, a short echo and a touch of room.
2	**Velvet Lead**	1/8D	Smooth lead vocal with a dotted eighth echo locked to tempo.
3	**Slapback Gold**	110 ms	Short vintage slapback for rockabilly, soul and upfront pop.
4	**Ping Pong Pop**	1/4	Quarter note echoes bouncing wide across the stereo field.
5	**Dream Halo**	520 ms	Big airy bloom for ambient vocals and pads of voice.
6	**Tape Echo 1/8**	1/8	Warm, wobbly tape-style eighth note repeats.
7	**Trap Triplets**	1/8T	Tight, bright triplet echoes for trap and modern hip-hop hooks.
8	**Afro Bounce**	1/8D	Dotted eighth groove echo with wide bounce for afrobeats.
9	**Soul Room**	180 ms	Warm, close room with a hint of grit for soulful lines.
10	**Jazz Club**	240 ms	Light compression, natural room and a soft echo for intimate jazz.
11	**Radio Telephone**	150 ms	Narrow, crunchy broadcast tone for lo-fi intros and effect verses.
12	**Underwater Mod**	420 ms	Deep chorus-style wobble with dark repeats.
13	**Vocal Lift**	1/4	Bright, forward vocal for choruses with a clean quarter note echo.
14	**Whisper Air**	300 ms	Breathy, sparkling top end for soft and close vocals.
15	**Cathedral**	600 ms	Huge stone-hall space with slow echoes for dramatic moments.
16	**Dub Throw**	1/4D	Long dotted quarter feedback trails for dub style word throws.
17	**Lo-Fi Cassette**	330 ms	Dusty, saturated and wobbly like an old four track.
18	**Hip-Hop Ad-Lib**	1/8	Wide eighth note throws that make ad-libs jump out of the mix.
19	**Neon Shimmer**	350 ms	Glittering fast modulation and a bright, glassy reverb.
20	**Wide Choir**	480 ms	Layered, swirling stereo repeats that sound like a stacked choir.

Recipes and tips

Lead vocal polish

Start with Velvet Lead. Set Low Cut near 90 Hz, De-Ess to taste, Squash around 50% and a little Drive. Keep Echo Mix under 25% so words stay clear.

Loop-station layers

Use Spread near 100% on background layers and a lower Mix on the lead. Layer presets such as Wide Choir and Dream Halo turn simple loops into stacked harmonies.

Making echoes sit behind the voice

Raise Lo Cut and lower Hi Cut in the Echo section. Thinner, darker repeats leave room for the lead vocal.

Ad-libs and throws

Use Hip-Hop Ad-Lib or Dub Throw, then automate Echo Mix in your DAW so only the last word of a phrase repeats.

Tape feel

Combine Drive at 40 to 50%, Mod Depth at 30 to 40% and Hi Cut near 3.5 kHz.

Gain staging

Squash adds automatic makeup gain. Check the level after changing it and use Output to match the bypassed level for fair comparisons.

Troubleshooting

Problem	Solution
Plug-in missing in Logic Pro	Restart Logic Pro, then use Settings, Plug-in Manager, Reset and Rescan Selection. Confirm VoxHalo.component sits in /Library/Audio/Plug-Ins/Components.
Plug-in missing in another DAW	Enable VST3 scanning in the DAW preferences and rescan.
macOS blocks the installer	Control-click the installer, choose Open, or use Open Anyway in Privacy and Security.
Echo timing drifts from the beat	Turn Sync on and make sure the DAW project tempo is set. Hosts that share no tempo give 120 BPM.
Feedback squeal in Standalone	Use headphones or lower the input level. Mute the monitor speakers.
Mono track	VoxHalo is a stereo plug-in. Place it on a stereo track or an aux send. Most hosts upmix a mono track automatically.

Technical specifications

Item	Detail
Formats	Audio Unit (effect), VST3, Standalone app

Item	Detail
Architecture	Universal (arm64 and x86_64), macOS 11 or later
Channels	Stereo in, stereo out
Latency	0 samples
Maximum echo time	2.9 seconds (tempo-synced times are clamped to this)
Reverb tail	Reported to the host as 6 seconds
Interface size	600 x 240 up to 2400 x 960 points, fixed aspect ratio
Presets	20 factory presets, host program list supported

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